

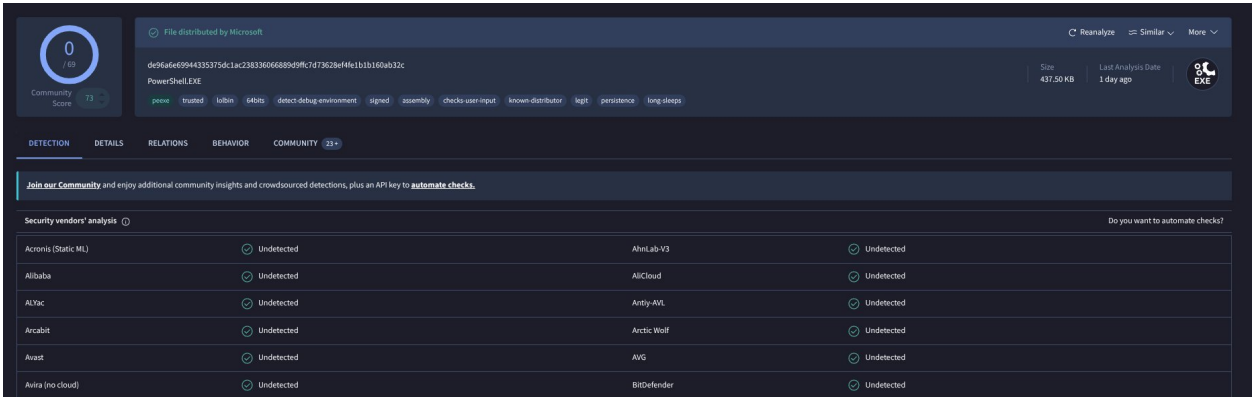
Event Information:

EventID :
64
Event Time :
Feb, 21, 2021, 07:23 PM
Rule :
SOC130 - Event Log Cleared
Level :
Security Analyst
Source Address :
172.16.20.3
Source Hostname :
Exchange Server
File Name :
powershell.exe
File Hash :
7353f60b1739074eb17c5f4dddefe239
File Size :
437.50 KB
Device Action :
Allowed

Analysis Steps:

Logs were searched with IP 172[.]16.20.3 (Exchange Server) as the source but nothing of note was found in network or email logs that might explain why Logs on this device were cleared, which is a red flag. The endpoint was also analysed via running services but there was nothing here either for the relevant time-frame that might explain why logs were cleared.

When searching for the MD5 hash on VirusTotal we can see that this is the genuine Powershell and not a malicious binary masquerading as it.



The screenshot shows the VirusTotal interface for a file named PowerShell.EXE. The file is identified as 'File distributed by Microsoft' with a SHA256 hash of de96a6e6994433575dc1ac23833666889d98c7d73628e4f0c1b1b160ab32c. The file size is 437.50 KB and it was last analyzed 1 day ago. The interface includes tabs for DETECTION, DETAILS, RELATIONS, BEHAVIOR, and COMMUNITY. A table titled 'Security vendors' analysis' shows results from 10 vendors, all of whom have detected the file as 'Undetected'.

Security vendor	Result
Acronis (Static ML)	Undetected
Alibaba	Undetected
ALYac	Undetected
Arcabit	Undetected
Avast	Undetected
Avg (no cloud)	Undetected
Avira (Static ML)	Undetected
Avira (no cloud)	Undetected
BitDefender	Undetected
BitDefender	Undetected

This activity must be determined as malicious until proven otherwise as it is the exact behaviour an attacker might engage in to remove evidence of malicious activities on a high-value corporate asset.

Based on following Playbook procedure and the analysis of available information, this alert is determined to be a successful True Positive.

Report:

Incident Summary

Analysis confirms a True Positive alert for Defense Evasion (Event Log Cleared) on the host Exchange Server (172[.]16.20.3). The alert was triggered by the execution of a legitimate Microsoft powershell.exe binary.

Incident Details

- **Alert Name:** SOC130 - Event Log Cleared
- **Source Host:** Exchange Server (172[.]16.20.3)
- **Flagged File:** powershell.exe
- **File Hash:** 7353f60b1739074eb17c5f4dddefe239
- **Device Action:** Allowed

Analysis and Evidence

- **Execution:** The alert indicates powershell.exe was utilised to clear system event logs.
- **Telemetry Gap:** A review of the endpoint and network logs for the relevant time-frame yielded no immediate evidence of prior compromise. However, the successful clearing of logs inherently destroys local forensic evidence.

Threat Intelligence

- **OSINT:** VirusTotal analysis of the file hash confirms it is a legitimate administrative tool, explicitly flagged as "File distributed by Microsoft" with a 0/69 detection rate. It is also tagged as a lolbin (Living Off The Land Binary), indicating it is a legitimate administrative tool frequently abused by threat actors to bypass security controls.

Actions Taken

1. Verified the binary as legitimate Microsoft PowerShell.
2. Reviewed available surrounding network and endpoint telemetry.
3. Declared the alert a True Positive based on the highly suspicious administrative action (log clearing) and escalated.

Note: LetsDefend expects the alert to be closed as a False Positive for full points.

Action taken 3. could have been:

3. Declared the alert as a False Positive based on a lack of clearly malicious activity but escalated due to highly suspicious administrative action (log clearing).

Let's Defend Result

SOC130 - Event Log Cleared

True Positive (-2 Point)

Analyze Malware (+5 Point)

Check if the malware is quarantined/cleaned (+5 Point)

My instinct would be to close this as True Positive as this must be followed up it would be reckless to close as a False Positive without knowing that it would be properly investigated. However, this activity is not certain to be malicious. If it were possible to close as False Positive and know that it would be investigated, this would perhaps be the most logical approach. Otherwise I would definitely close it as a True Positive.

I think the culture of log clearing would be specific to an organisation in terms of how often it is performed directly by staff but this could be an unhealthy culture as log clearing is an activity attackers use when covering tracks. It would make sense that if this must be done that it should be communicated and if there is no communication then an alert should be escalated.

Closing as False Positive gave full points.

SOC130 - Event Log Cleared

False Positive (+5 Point)

Analyze Malware (+5 Point)

Check if the malware is quarantined/cleaned (+5 Point)